INSTALLATION & MAINTENANCE MANUAL

FOR

ELECTRO HYDRAULIC DOCK LEVELER

CLIENT : - SRD

PROJECT : - SINNER (NASHIK)

MFG.MONTH & YEAR : - JUNE 2019.

DESIGNED, DEVELOPED & MARKETED,

BY,

NIHVA TECHNOLOGIES PVT. LTD
G-BLOCK, 1ST FLOOR, THERMAX CHOWK
MIDC, ABOVE RENAULT SHOWROOM
CHINCHWAD, PUNE-411019

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A. DESPATCH & UNLOADING: -

1. Loading, for dispatch, will be our responsibility. On receipt of the Dock-leveler at site, please check and ensure that there is no physical damage to the equipment.

2. While unloading the equipment from the transporting vehicle, please ensure to use slings with rated capacity higher than the gross weight of the Stationary lift in packed condition.

3. Slings MUST be hooked ONLY at specified positions. These positions are indicated and marked with a figure of 'HOOK'. Improper sling position or improper slinging could cause damage to the equipment & serious accidents.

B. ORDERING OF SPARES: -

1. When ordering for spare, please specify the following in your Purchase Order.

   - Equipment Name
   - Model Number
   - Equipment Serial No.
   - Part name / Description
   - Part Number
   - Quantity required
   - Delivery period / time
   - Mode of dispatch

   This would avoid ambiguity and help us serve you better.

DATASHEET FOR DOCK LEVELER:-

CLIENT: - IKEA, CHAKAN PHASE III

1.0 CONSTRUCTION DETAILS

1.1 Required Capacity - each	: 6 ton
1.2 Model No.	: NDL
1.3 Leveler operation mode	: Hydraulically with AC operated.
1.4 Lip operation mode	: hydraulic
1.5 Operation control	: by push buttons and control panel
1.6 Loading platform height from road Level	: 4 Feet (Will be confirmed during Detailing)
1.7 Travel above & below dock level	: (+) 300 / (-) 200 mm
1.8 Platform Size (L x W)	: 2250 x 2025 mm
1.9 Lip size - (L x W)	: 2025x 400 mm
1.10 Working length	: 2250 mm
1.11 Overall length	: 2650 mm
1.12 Maximum slope can be achieved	: 8 to 10%
1.13 Clear Pit dimensions (L x W X D)	: 2250 x 2050 x 485 mm
1.14 Hydraulic power pack	
1.14.1 Driver type	: Direct by electric Motor
1.14.2 Absorbed power	: 0.75 kW
1.14.3 Motor Ratings	: 0.75/1 kW /HP
1.14.4 Degree of protection for motor	: IP 55
1.14.5 Class of Insulation for motor	: Class F with temp. Rise limited to Class B
1.14.6 Motor RPM	: 1410
1.14.7 Pump design	: 1.6 cc/rev.
1.14.8 Pump pressure	: 200 (max) Kg/cm²g
1.14.9 Safety valve set pressure	: 90 Kg/cm²g
1.14.11 Pump type	: Gear
1.14.12 Valves required	: By vendor
1.14.13 Volume of oil receiver	: 5 Lit
1.14.15 First fill of oil / Qty	: By Vendor
1.14.16 Hydraulic tubing	: By Vendor
1.14.17 Instrumentation	: By vendor
1.14.18 Control panel	: By vendor
1.15 Empty weight	: 1040 Kg

DOCK LEVELLER

INTRODUCTION: -

The hydraulic Dock Leveler is the result of many years' experience in materials handling engineering. "NIHVA" offers the best possible factor of safety with simple, easy to use controls. The high quality of manufacture gives a robust construction and many years of useful life. Each lip and platform movement can be individually controlled (or corrected) by push button.

The dock leveler can be set at any position. During loading, the Dock Leveler is in floating mode i.e. it follows the up/down movements of the vehicle deck. Even with lopsided vehicles, the Dock leveler lip will remain in contact, and toe-traps are avoided by the tilt built into the Dock Leveler.

The whole platform is designed to flex laterally adequately to adapt to sideways tilted vehicles. The leveler always seals perfectly to the lorry bed.

PRINCIPLE: -

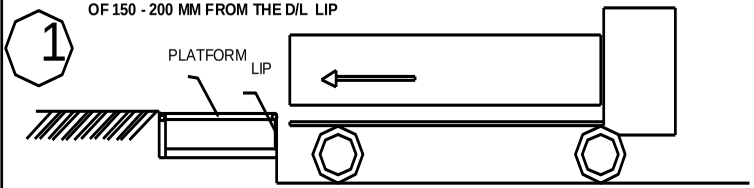
Dock Leveler is an adjustable platform used between the transport vehicle and dispatch area of the warehouse of factory for efficient and safe material transfer applications. It bridges the gap between the dispatch area and the transport vehicle and compensates for variations in the truck bed height.

PRE – OPERATING INSTRUCTIONS: -

1. Ensure Hydraulic Oil in the power pack.
2. Ensure Motor connection and proper earthing.
3. Ensure joint leakage.
4. Ensure proper location of platform and lip.

1

REVERSE THE TRUCK UPTO DISTANCE
OF 150 - 200 MM FROM THE D/L LIP

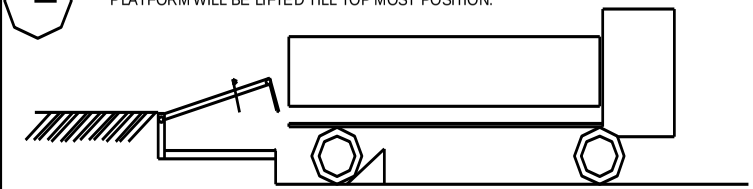


NOTE : ENSURE THAT EDGES OF TRUCK LOAD BODY ARE OUTSIDE LIP WIDTH
I.E. CHECK ALIGNMENT OF LIP WITH TRUCK BODY.

2

HOLD PLATFORM UP BUTTON IN PRESSED
POSITION.

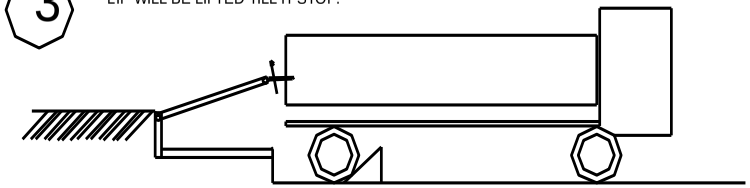
PLATFORM WILL BE LIFTED TILL TOP MOST POSITION.



3

HOLD LIP UP BUTTON IN PRESSED POSITION.

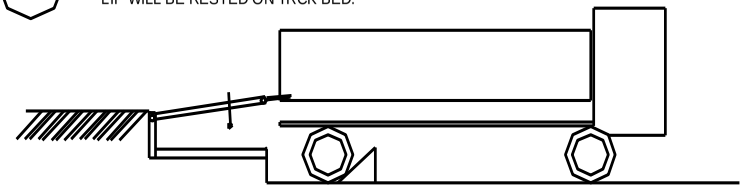
LIP WILL BE LIFTED TILL IT STOP.



4

HOLD PLATFORM DOWN BUTTON IN PRESSED
POSITION.

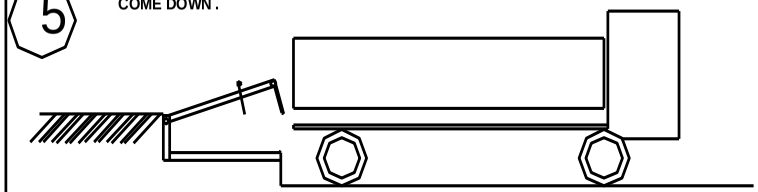
LIP WILL BE RESTED ON TRCK BED.



UNLOAD OR LOAD THE MATERIAL FROM THE TRUCK.

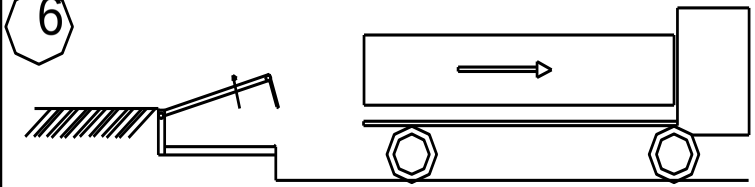
5

PRESS PLATFORM UP TO CLEAR THE LIP
FROM TRUCK BED THAT TIME THE LIP WILL
COME DOWN .



6

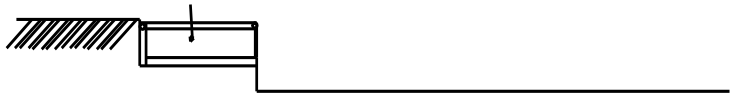
DRIVE THE TRUCK AWAY FROM D/L.



7

HOLD PLATFORM DOWN BUTTON .

LIP WILL BE PARKED TO ITS ORIGINAL POSITION.



OPERATING INSTRUCTION FOR DOCK- LEVELER.

NIHVA TECHNOLOGIES PVT. LTD.

ROUTINE MAINTENANCE AND LUBRICATION: -

These recommendations are applicable to the Dock Leveler considering average daily usage of 15 to 20 times. If it is used more frequently, the same time intervals for the maintenance should be adjusted accordingly.

DAILY: -

1. Check Hydraulic pipe lines for leak proof working.
2. Check electrical connections for their proper working.

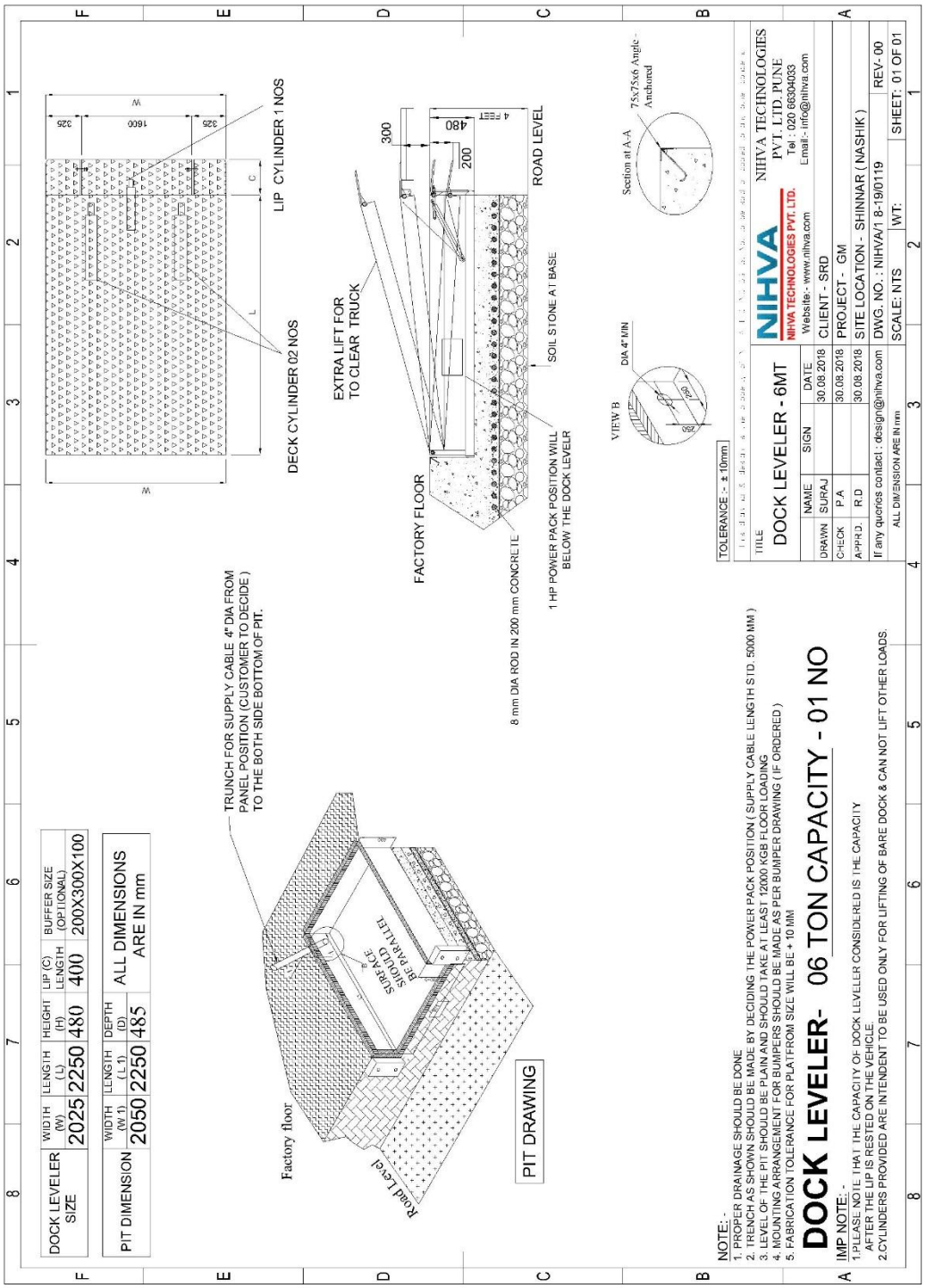
MONTHLY: -

1. Grease the hinge pins at bottom frame and top plate joint and extension lip plate hinge joint.
2. Grease the main Jack Cylinder and Lip Jack Cylinder, Top and Bottom pins if some frictional sound hears.
3. Change Hydraulic Oil (servo 68) viscosity from hydraulic expert & change if required.

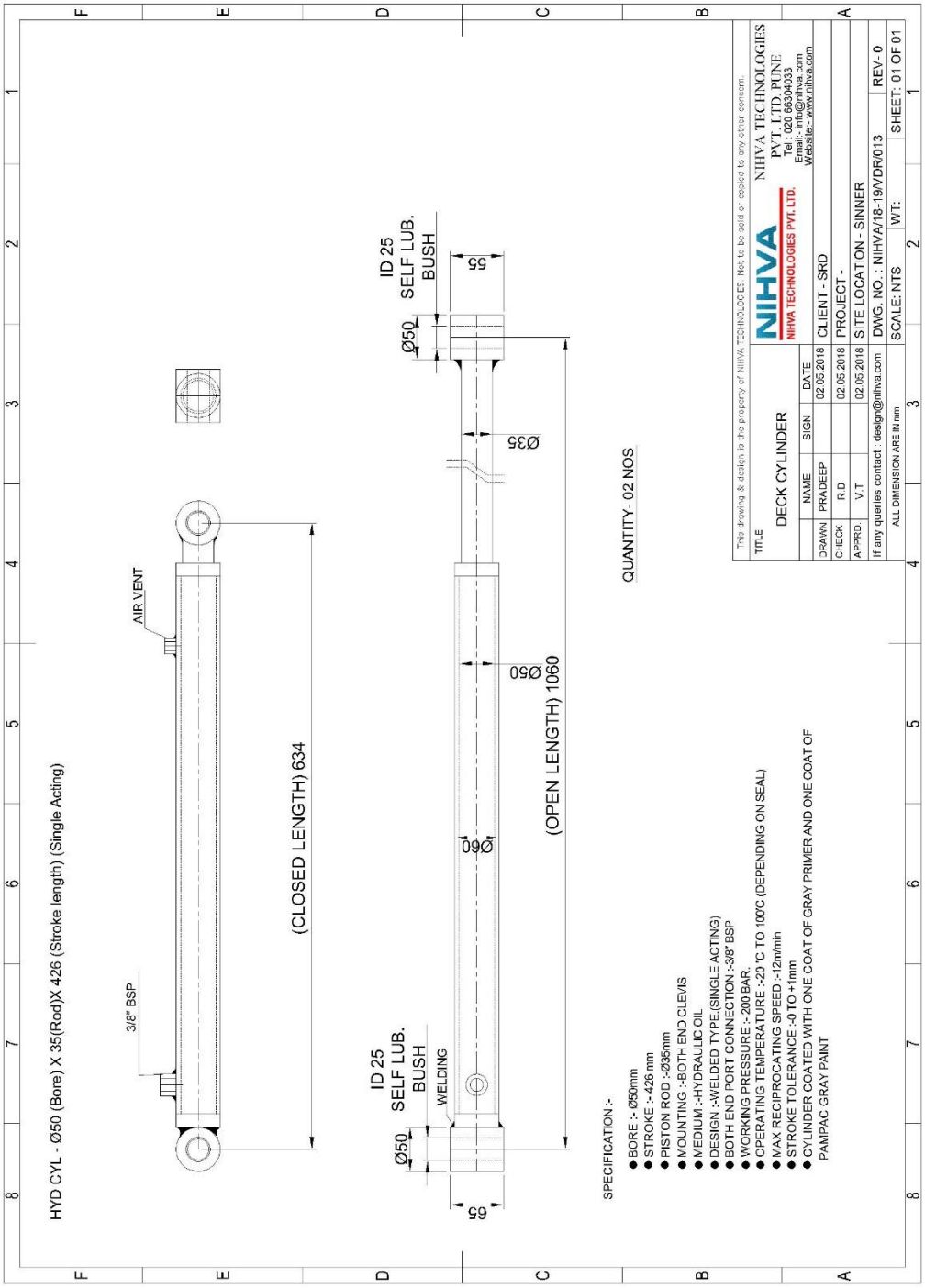
ASSEMBLIES & SPARE PARTS LIST: -

1. GA DRAWING.
2. DECK CYLINDER DETAILS.
3. LIP CYLINDER DETAILS.
4. HYDRAULIC POWER PACK DETAILS & CIRCUIT.
5. ELECTRICAL CIRCUIT.

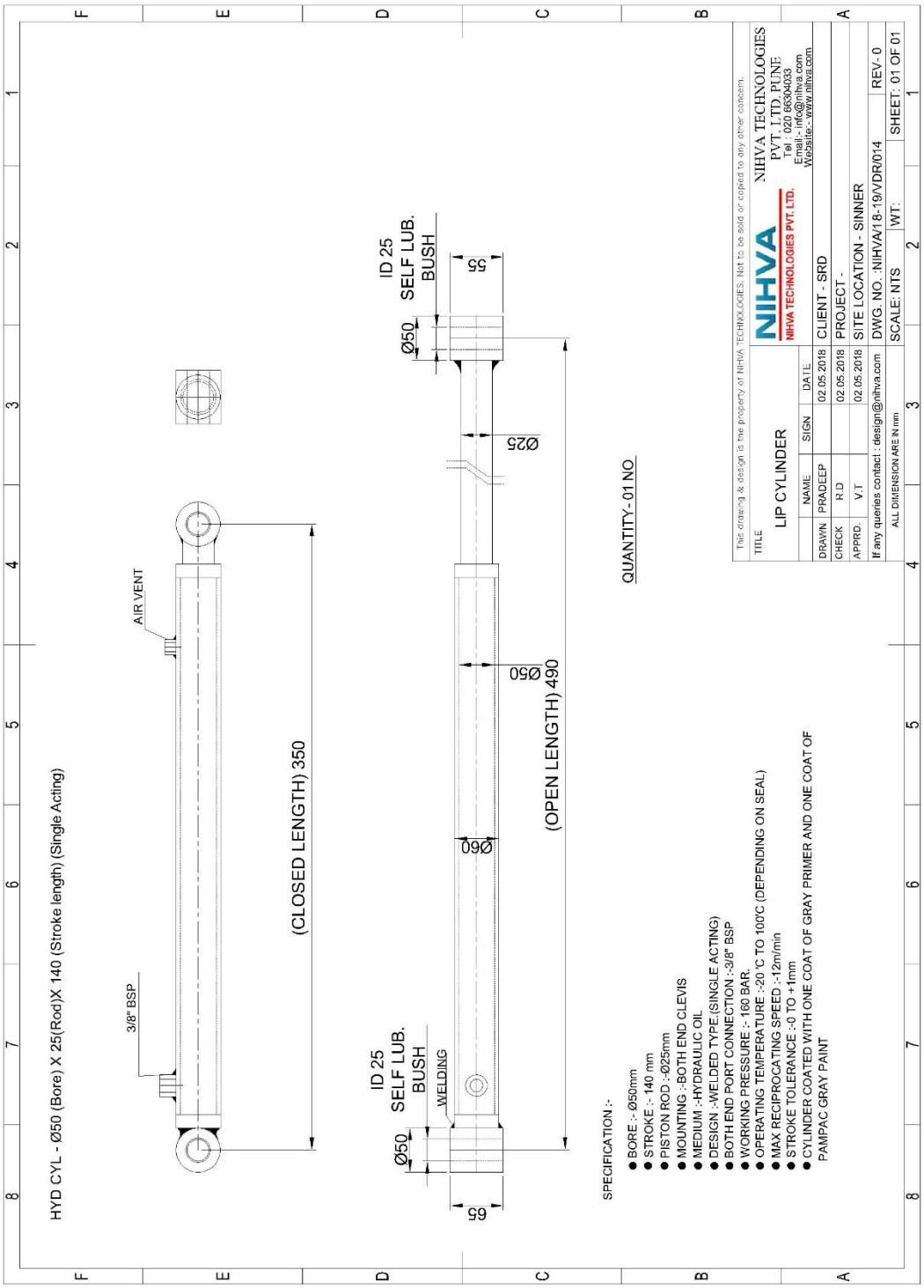
GA DRAWING:-


DOCK LEVELER SIZE
WIDTH (W) 2025
LENGTH (L) 2250
HEIGHT (H) 480
LIP (C) LENGTH 400
BUFFER SIZE (OPTIONAL) 200X300X100
PIT DIMENSION
WIDTH (W 1) 2050
LENGTH (L 1) 2250
DEPTH (D) 485
ALL DIMENSIONS ARE IN mm
TRUNCH FOR SUPPLY CABLE 4" DIA FROM PANEL POSITION (CUSTOMER TO DECIDE) TO THE BOTH SIDE BOTTOM OF PIT.
LIP CYLINDER 1 NOS
DECK CYLINDER 02 NOS
EXTRA LIFT FOR TO CLEAR TRUCK
FACTORY FLOOR
ROAD LEVEL
SOIL STONE AT BASE
1 HP POWER PACK POSITION WILL BELOW THE DOCK LEVELR
8 mm DIA ROD IN 200 mm CONCRETE
Section at A-A
75x75x6 Angle - Anchored
VIEW B
DIA 4" M/N
Factory floor
Road Level
SURFACE SHOULD BE PARALLEL
PIT DRAWING
TOLERANCE :- ± 10mm
TITLE
DOCK LEVELER - 6MT
NIHVA TECHNOLOGIES PVT. LTD. PUNE
Tel : 020 66304033
Email :- info@nihva.com
Website:- www.nihva.com
CLIENT - SRD
PROJECT - GM
SITE LOCATION - SHINNAR (NASHIK)
DWG. NO. : NIHVA/1 8-19/0119
REV- 00
SCALE: NTS
WT:
SHEET: 01 OF 01
NOTE: -
1. PROPER DRAINAGE SHOULD BE DONE
2. TRENCH AS SHOWN SHOULD BE MADE BY DECIDING THE POWER PACK POSITION (SUPPLY CABLE LENGTH STD. 5000 MM)
3. LEVEL OF THE PIT SHOULD BE PLAIN AND SHOULD TAKE AT LEAST 12000 KGB FLOOR LOADING
4. MOUNTING ARRANGEMENT FOR BUMPERS SHOULD BE MADE AS PER BUMPER DRAWING (IF ORDERED)
5. FABRICATION TOLERANCE FOR PLATFROM SIZE WILL BE + 10 MM
DOCK LEVELER- 06 TON CAPACITY - 01 NO
IMP NOTE: -
1.PLEASE NOTE THAT THE CAPACITY OF DOCK LEVELLER CONSIDERED IS THE CAPACITY
AFTER THE LIP IS RESTED ON THE VEHICLE
2.CYLINDERS PROVIDED ARE INTENDENT TO BE USED ONLY FOR LIFTING OF BARE DOCK & CAN NOT LIFT OTHER LOADS.

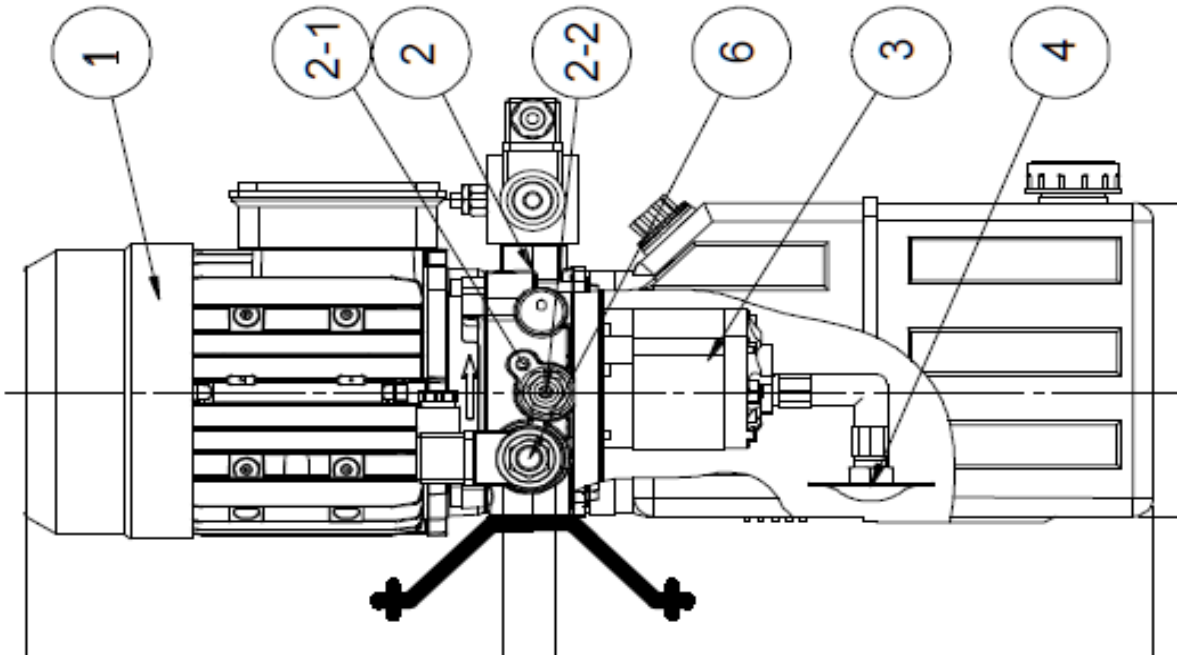
DECK CYLINDER: -



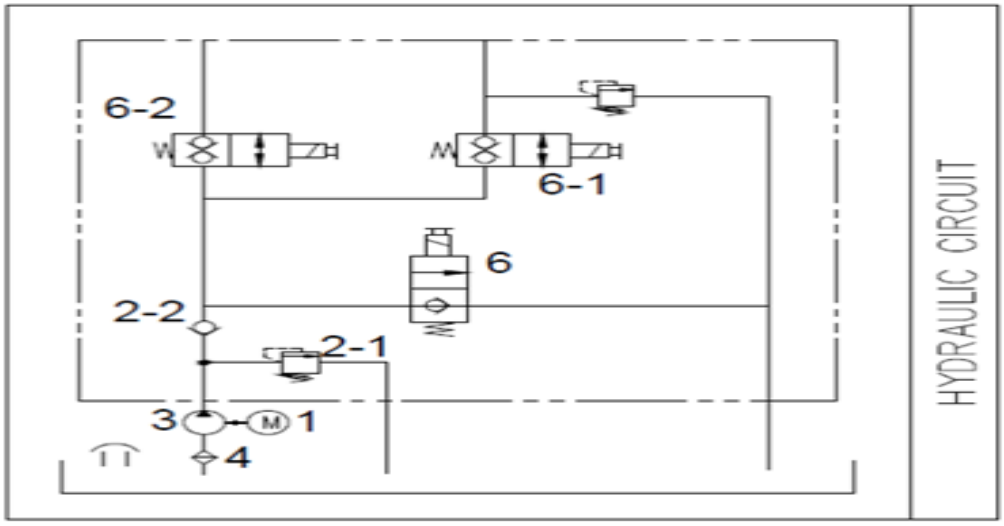
LIP CYLINDER :-



HYDRAULIC POWER PACK DETAILS: –



HYDRAULIC CIRCUIT:-



P & T PORT : 1/4" BSP
COIL : AC 220 V,

HYDRAULIC CIRCUIT PART LIST:-

SR NO	DESCRIPTION	QTY
1	MOTOR – 1 HP, 3 PHASES	01 NO
2	CHECK VALVE	01 NO
2.1	RELIEF VALVE (RF 1)	01 NO
2.2	RELIEF VALVE (RF 2)	01 NO
3	GEAR PUMP	01 NO
4	SUCTION STRAINER	01 NO
5	OIL TANK (5 LTR)	01 NO
6	SOLENOID COIL (220 V)	03 NOS

ELECTRIC CIRCUIT & PART LIST: -

SR.	DESCRIPTION.	PART NO.	QTY.
SV1	SOLENOID VALVE MAIN JACK	10000336	1
SV2	SOLENOID VALVE LIP JACK	10000336	1
SV3	SOLENOID VALVE LIP JACK	10000336	1
L1,L2 L3	INDICATOR LAMPS 220V	10000337	3
K1	CONTACTOR COIL 230V,AC]	10000338	1
OL	OVERLOAD RELAY (1,25A-2A)	10000339	1
SW1	MCB 3 POLE[10 A,440 V-24V]	10000340	1
P.B	PUSH BUTTON GREEN	10000659	1
P.B	PUSH BUTTON YELLOW	10000660	1
P.B	PUSH BUTTON RED	10000753	1
P.B	PUSH BUTTON BLUE	10000909	1

DOCK LEVELER 3 SOLENOID VALVE AC PANEL

